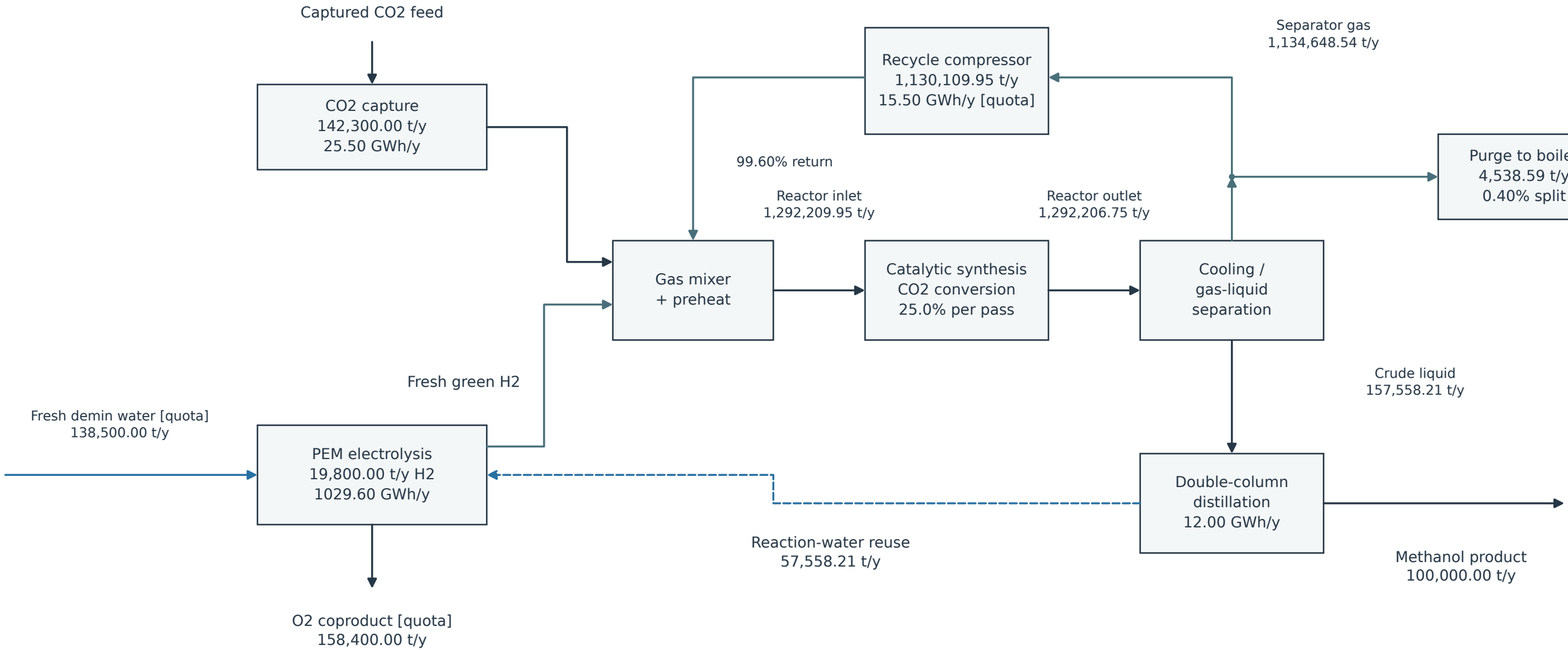
100,000 t/y green methanol: material and electricity flows

Actual task3 anchor B | Renewable share 100.0% | Purge 0.40% | Carbon utilization 96.5221%



Annual electricity: 1,082.60 GWh | PEM 95.10% | CO2 capture 2.36% | Recycle 1.43% | Distillation 1.11%

Basis: pure-component synthesis model; [quota]: source engineering values. Dashed blue: internal water reuse; widths are schematic.

PEM supply = fresh quota + modelled reuse = 196,058.21 t/y; H2 + O2 quota = 178,200.00 t/y.

Unaccounted PEM mass: 17,858.21 t/y. Missing outlet information is not assigned to wastewater or another invented flow.

Cooling make-up (policy model): 2,000.00 t/y; total fresh withdrawal: 140,500.00 t/y. Reuse is an internal flow.

Synthesis C/H/O checks passed. Source molecular-weight rounding leaves 3.195 t/y mass residual; PEM quotas are not fully closed.